

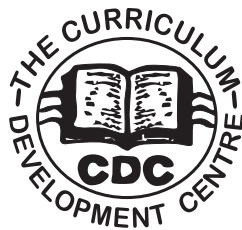


Republic of Zambia
Ministry of Education

GEOGRAPHY

TEACHING MODULE

FORM 1 - TERM 2



Developed by the Curriculum Development Centre
Lusaka
2025

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VISION

Quality, life- long education for all which is accessible, inclusive and relevant to individual, national and global needs.

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PREFACE

The introduction of the competence-based curriculum (CBC) marks a transformative step in improving education quality and relevance. This Form 1 Geography Teaching Module has been meticulously designed to support teachers to effectively deliver Geography lessons under the new curriculum. The module aims to bridge the gap created by the absence of approved textbooks and to equip educators with practical tools, activities, and assessments tailored to the learners' level and needs. As a Ministry, we understand the challenges that come with transitioning to a new curriculum. This module reflects our commitment to making this journey as smooth as possible for teachers, by providing structured guidance, suggested activities, and assessments that align with the CBC. However, we also encourage educators to supplement this module with their own research and innovations to address any gaps that may arise during implementation. The suggested activities and assessments provided in this module are designed to foster practical and critical thinking skills in learners while ensuring relevance and adaptability to different school contexts. Teachers are encouraged to tailor these suggestions to their unique classroom dynamics and to explore alternative acceptable approaches when necessary. This flexibility is key to ensuring that all learners, regardless of their environment, benefit from meaningful and impactful Geography education. We hope this module will serve as a valuable resource for teachers as they navigate the new Competence-Based Curriculum, enhancing their ability to inspire and equip learners with essential Geography knowledge, skills and values.



Kelvin Mambwe (PhD)
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ACKNOWLEDGEMENT

This Geography Teaching Module is designed to provide the teacher with a scope and sequence of topics considered necessary to be offered at Ordinary Secondary School Level in Form 1 of term II. This is with a view to provide guidance to the teaching and learning of this unique, but yet exiting blend of concepts in Geography for teachers and other experts in the field to appropriately offer relevant lessons at Ordinary Secondary Level.

Many thanks go to individuals, institutions and organisations that participated in the successful development of this module. These include; teachers, lecturers from colleges, public universities in Zambia and in particular, the University of Zambia.

Sincere gratitude also goes to the Zambia Education Enhancement Project (ZEEP) for the financial support and Zambia Educational Publishing House (ZEPH) for the technical support towards the development and finalisation of the module.

Last but not the least, the commitment and hard work of all the staff at the Curriculum Development Centre in ensuring that this module comes to reality is recognised.



Charles Ndakala (Dr.)
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INTRODUCTION

This Geography Teaching Module provides guidance to the teacher on how to deliver competence-based lessons on the fundamental concepts of Geography. This module aims to help a teacher develop an understanding of the indoor and outdoor environment required to prepare lessons which help learners develop critical thinking, problem-solving, and analytical skills. This module is committed to providing an enriching and supportive educational environment where teachers can prepare and deliver lessons which stimulate interest in Geography for learners. This module seeks to support teachers to prepare lessons for both academic performance and their future roles as knowledgeable and responsible citizens in a scientifically advanced society by encouraging inquiry, curiosity, and a love of the subject.

It is expected that this Geography teaching module, once implemented successfully, will inspire and empower learners to achieve their full potential, equipping them with the knowledge and skills necessary to navigate and contribute to the world around them. This module is designed to ensure learners develop a deep understanding of Geography principles while also acquiring the practical skills and competencies needed for further education and careers in Geography.

How to use the Module

To effectively use this Geography Teaching Module for Form 1:

- ☐ Read and familiarise yourself with the module's content, learning activities, and assessment guides.
- ☐ Plan your lessons in advance using the module's suggested teaching and learning activities.
- ☐ Use a variety of teaching methods including demonstrations, discussions, group work, and hands-on experiments.
- ☐ Encourage active learning by asking open-ended questions, promoting critical thinking, and fostering problem-solving skills.
- ☐ Assess learners learning regularly using the module's suggested assessment strategies and tools.
- ☐ Provide feedback and support to learners, helping them to identify areas for improvement and develop their skills.
- ☐ Integrate technology into your teaching using multimedia resources and interactive simulations to enhance student engagement and understanding.
- ☐ Monitor student progress and adjust your teaching strategies as needed, to ensure that all students meet the learning objectives

By following these steps, you can effectively use this Geography Teaching Module for Form 1 to support your teaching and promote learning.

Suggested Teaching and Learning Materials

The study of Geography content requires hands-on experiences, visual aids, and interactive resources to strengthen deep understanding and appreciation. To support the teaching and learning of Geography in Form 1, the suggested teaching and learning materials are meant to:

- ☐ Enhance learner engagement and motivation in Geography
- ☐ Develop practical skills and techniques in Geography
- ☐ Promote critical thinking, problem-solving, and analytical skills
- ☐ Support differentiated instruction and inclusive learning

By utilising these suggested teaching and learning materials, teachers can create a dynamic and supportive learning environment that promotes academic excellence, creativity, and scientific literacy in Geography for Form 1 learners.

Learning Environment Set Up

Creating an effective learning environment for teaching and learning Geography is important in deepening understanding and application of concepts in real life context. The learning environment set-up aims to create a safe, inclusive, and engaging space.

- ☐ Indoor learning set up: Conducted in a classroom with controlled environment including seating arrangements, lighting and temperature control.
- ☐ Outdoor learning set up: Takes place in open spaces such as school surroundings, gardens, or natural environments

Teaching Methodology

The effective teaching methodologies in Geography include:

- ☐ Conducting experiments: Demonstrate key principles and encourage curiosity among learners.
- ☐ Collaborative learning: Pair learners to work together, promoting peer-to-peer teaching, discussion, and problem-solving.
- ☐ Conceptual learning: Connect Geography concepts to everyday life, industry, or current events, making learning relevant and meaningful.
- ☐ Differentiated instructions: Tailor teaching to meet diverse learning styles, abilities, and interests of different learners.
- ☐ Feedback and Reflection: Encourage learners to reflect on their learning, providing constructive feedback to guide improvement.
- ☐ Inquiry-based learning: Encourage learners to explore, investigate, and discover Geography concepts through hands-on experiments and activities.
- ☐ Integration of Technology: Use digital tools, simulations, and visualisations to enhance engagement, understanding, and analysis.
- ☐ Problem-based learning: Present real-world problems or case studies, requiring students to apply Geography principles to develop solutions.
- ☐ Project-based learning: Assign open-ended projects, allowing students to design, conduct, and present research or applications of Geography concepts.

By implementing these methodologies, a teacher can create an engaging, inclusive, and effective learning environment.

Time Allocation

The standard minimum learner-teacher contact time for Geography at Secondary School Level is 3 hours 20 minutes per week, translating into five (5) periods. The duration for a single period is 40 minutes. The contact time at Secondary School Level is planned in such a way as to give ample time for practical activities.

1.1. TOPIC 1: NATURAL RESOURCES

Introduction

Natural resources are materials like water, air, trees, and minerals that come from nature and are essential for life. Some of these such as sunlight and forests, can be replaced over time, while others, such as oil and minerals, take millions of years to form and could run out.

Responsible use of natural resources is important to ensure they last. This includes avoiding waste, protecting the environment, and using resources wisely. As Zambia's population grows, more resources are needed for food, energy, and development. This creates pressure on natural resources, so careful management is necessary to balance today's needs with future sustainability. Managing natural resources involves actions like replanting trees, protecting water sources, and reducing pollution. By doing this, we can support both the people of today and future generations while keeping the environment healthy.

This topic will focus on exploring responsible use of natural resources, the relationship between population and these resources and sustainable management practices that ensure their preservation for future generations while meeting the needs of the present.

Hook: *'Have you wondered where your desk, clothes and the food you eat (sandwich, nshima, bread) comes from? The answer lies in something we often take for granted: natural resources'*

General Competences

The general competencies in this topic are: Analytical thinking, Citizenship, Collaboration, and Environmental Sustainability.



Key Terms:

Key terms

- ☐ Natural Resources
- ☐ Preservation
- ☐ Sustainable management
- ☐ Practices
- ☐ Carrying Capacity

SUB-TOPIC 1.1.1: NATURAL RESOURCES

Natural resources are things found in nature that people use to live and grow. They include:

Renewable resources such as sunlight, water, and trees, which can be replaced naturally over time and **non-renewable resources** such as oil, coal, and minerals, which take a long time to form and can run out. These resources help us build homes, make electricity, grow food, and more. Managing them carefully ensures they last for future generations. This sub-topic will explore responsible use of natural resources.

Specific Competence (s) ;

Learners to -:

Use renewable and non-renewable resources responsibly



Activity 1

Identifying different natural resources

Learning Environment

The learning activity would be outdoor or indoor set up.

Indoor Set up

- ☐ Divide learners into smaller groups based on their strengths to foster collaboration.
- ☐ Ask each group to brainstorm and list environmental or natural resources they are familiar with.
- ☐ Provide charts, pictures, or physical representations of natural resources to enhance visual learning.
- ☐ Encourage discussion within the groups guiding them to explore examples such as water, forests, air, soil, minerals and wildlife

Outdoor Set up:

- ☐ Before heading outdoors, discuss with learners what natural resources are and provide examples of both living (e.g., plants, animals) and non-living (e.g., rocks, soil, water) resources.

Observation Walk:

- ☐ Take the learners to an outdoor area, such as a school surrounding, nearby park, or community space.
- ☐ Ask them to observe their surroundings and identify as many natural resources as possible in the environment.

Categorization:

- ☐ Divide learners into small groups and have each group categorize the resources they identified into living and non-living.
- ☐ Encourage discussions within the groups to refine their understanding.

Presentations:

- ☐ Let each group share their findings with the class in a brief presentation. They can use sketches, diagrams, or other creative methods to illustrate their observations.

**Content Tips**

- ☐ Natural resources are the materials and substances that occur naturally within the environment and are used by humans to fulfill their needs and wants.
- ☐ There are two types of natural resources and these are renewable and non-renewable resources.
- ☐ These resources are essential for sustaining life and supporting economic activities.
- ☐ Identifying natural resources involves understanding the types of resources available in nature and the methods used to locate, classify, and evaluate them.
- ☐ They include;
 - ✓ Water resources such as Rivers, lakes, groundwater, rainwater, etc.
 - ✓ Forests resources such as trees, medicinal plants, bamboos.
 - ✓ Energy resources such as solar, wind, hydro, geothermal, biomass, fossil fuels.
 - ✓ Soil and agricultural resources
 - ✓ Fossil fuels (Coal, Oil, Natural Gas)
 - ✓ Minerals and metals such as copper, gold,
- ☐ Methods of identifying natural resources include the following
 - ✓ Geological surveys
 - ✓ Remote sensing

- ✓ Exploratory drilling
- ✓ Sampling and analysis
- ✓ Field surveys
- ✓ Lab analysis confirms composition and quality of resources.
- ☐ Importance of identifying natural resources
 - ✓ Economic benefits; through extraction process and trade.
 - ✓ Sustainable management; understanding the distribution, accessibility and quality of natural resources can inform sustainability management practices.
 - ✓ Environmental conservation; it helps to mitigate the environmental impacts of the extraction and use.
- ☐ Challenges in identifying natural resources
 - ✓ Technological limitations; such as accessing remote or hard-to-reach areas.
 - ✓ Lack of knowledge
 - ✓ Social and economic factors; such as local communities, government policies, and market demand.

Teaching and Learning Materials

- ☐ Note books for outdoor activities.
- ☐ Pictures and videos of Natural resources
- ☐ Samples of some natural resources



Assessment

- ☐ Learners through discussions, debates demonstrate how some resources could be depleted.
- ☐ Learners explain the dangers of depleting the natural resources.
- ☐ Learners display posters on how to avoid depletion of natural resources.
- ☐ Learners plant trees as a means of avoiding forestry depletion



Activity 2

Describing the types of natural resources: (*Renewable - water, land, wildlife, solar... and non-renewable - minerals, fossil fuels...*)

Learning Environment

The learning activity would be outdoor or indoor set up.

Indoor Set up:

- ☐ Divide learners into smaller groups, ensuring diversity within each group to encourage collaboration and varied perspectives.
- ☐ Check learner understanding of concepts of renewable and non-renewable resources. Provide clear definitions and examples:
 - **Renewable resources:** Those that replenish naturally and can be used repeatedly (e.g., sunlight, wind, water).
 - **Non-renewable resources:** Those that are finite and cannot be replaced after use (e.g., fossil fuels, minerals).
- ☐ Have each group discuss and classify the listed resources into **renewable** and **non-renewable** categories.
- ☐ Encourage learners to justify their categorization by explaining their reasoning.
- ☐ Let each group present their findings creatively, such as:

- Drawing posters showing renewable and non-renewable resources.
- Preparing short skits or role-plays illustrating the usage of these resources.
- ❑ After presentations, hold a group discussion to emphasise:
 - Why conserving non-renewable resources is important.
 - How renewable resources can be maximized.

Outdoor Set up:

- ❑ Ask learners to observe and describe the natural resources found in their immediate environment.
- ❑ Learners to compile their observations of natural resources and categorise them into renewable and non-renewable describe Renewable and non- renewable resources.



Content Tips

- ❑ **Renewable resources** are natural resources that can replenish or regenerate over time through natural processes or sustainable management practices.
- ❑ These resources are typically derived from the Earth's natural cycles, such as the water cycle, carbon cycle and photosynthesis, ensuring they are available for continuous use if managed responsibly.
- ❑ Examples of renewable resources include sunlight, wind, water and forests.
- ❑ Renewable resources can be continuously reproduced or regenerated, ensuring their availability for future generations if managed sustainably.
- ❑ **Non-renewable resources** are those that form over a long period of time and cannot be replaced once depleted.
- ❑ Examples include fossil fuels (coal, oil, natural gas) and minerals (gold, copper, iron), etc.
- ❑ Non-renewable resources are finite and their extraction and use often lead to environmental degradation and resource depletion.

Teaching and Learning Materials

- ❑ Note books for outdoor activities.
- ❑ Pictures and videos of Natural resources
- ❑ Samples of some natural resources



Assessment

- ❑ Learners explain the importance of renewable and non-renewable resources
- ❑ Learners display posters on how to avoid depletion of natural resources.



Activity 3

Utilizing Renewable and Non-renewable Resources responsibly

Learning Environment

- ❑ The learning activity would be indoor or indoor set up.

Context Setting

- ❑ Arrange desks in small groups of 4–6 learners to encourage collaboration.
- ❑ Brief learners on the importance of using natural resources responsibly, emphasising the difference between renewable (such as solar energy and water) and non-renewable resources (such as fossil fuels and minerals).

- ☐ Use visual aids or examples to illustrate how responsible or irresponsible practices impact the environment and future generations.

Identifying Responsible Activities

- ☐ Assign each group to brainstorm and list activities that promote responsible use of natural resources. Examples might include recycling, water conservation, using public transport, or planting trees.
- ☐ Ask learners to write down their ideas on sticky notes or chart paper.
- ☐ Groups should discuss how these activities benefit the environment and support sustainability.

Identifying Irresponsible Activities

- ☐ Groups then identify and list activities that represent irresponsible use of natural resources, such as littering, overfishing, deforestation, or excessive fossil fuel consumption.
- ☐ Encourage learners to analyze why these activities are harmful and their long-term impacts.

Discussion

- ☐ Groups present their findings and solutions to the class using visual aids.
- ☐ Summarize the key takeaways, highlighting responsible practices learners can apply in their daily lives.

Outdoor Set up:

Observation Walk:

- ☐ Select areas within the school premises or the local environment that showcase examples of resource use (e.g., water taps, gardens, energy sources, waste disposal sites).
- ☐ Groups explore the school or local environment and document examples of responsible (e.g. water conservation, recycling) and irresponsible practices (e.g., littering, deforestation).
- ☐ Encourage learners to engage with the environment actively, noticing patterns and asking questions such as “How is this resource being used, and what impact does it have?”

At-Risk Resources

- ☐ Groups identify specific resources that are at risk due to irresponsible practices (e.g., polluted water sources, eroded soil).
- ☐ Learners discuss the potential consequences of continued misuse, connecting their observations to larger environmental challenges.
- ☐ Each group brainstorms practical solutions to mitigate irresponsible practices and protect at-risk resources. Examples might include planting trees, educating peers on resource conservation, or setting up recycling stations.

Discussion

- ☐ Groups present their findings, risks identified, and proposed solutions to the class.
- ☐ Encourage collaboration to refine ideas further.

**Content Tips**

- ❑ Utilizing renewable and non-renewable resources refers to the process of extracting, managing, and using these natural resources to meet human needs while considering their availability and the impact of their use on the environment and future generations.
- ❑ It involves harnessing them for energy, materials, or other purposes in a way that maintains their ability to regenerate. For instance:
 - ✓ Generate electricity using water, solar, wind, geothermal and converting organic materials like wood, crop and animal waste, or algae.
 - ✓ Materials such as wood for construction, minerals for manufacturing and processing purposes and fossil fuels for energy production.
 - ✓ Other materials such as clay, sand and gravel are used for construction, building and other applications.
- ❑ Responsible use of renewable resources means;
 - ✓ Investing in clean technologies to optimize renewable resource use; this involves employing advanced systems, processes, and innovations that minimize environmental impact, enhance efficiency, and promote sustainability. Using clean technologies not only harnesses renewable resources efficiently but also minimizes ecological damage, supports economic development, and builds resilience against climate change.
 - ✓ Avoid over-exploitation, such as over extraction of minerals, excessive logging or water diversion.
 - ✓ Minimize environmental impacts, such as habitat disruption or pollution.
 - ✓ Promote resource conservation and restoration.
- ❑ Responsible use of non-renewable resources means;
 - ✓ Minimizing consumption to ensure that they last long.
 - ✓ Maximize efficiency to reduce waste and extend the lifespan of reserves.
 - ✓ Implement recycling programs to recover and reuse materials.
 - ✓ Transition to alternative, renewable energy sources to reduce reliance on non-renewable natural resources.
 - ✓ Mitigate environmental damage through stricter regulations and remediation efforts.
- ❑ Irresponsible use of renewable and non-renewable resources
 - ✓ Cutting down of trees for various purposes without replacing them
 - ✓ Overfishing and use of bad methods of fishing
 - ✓ Overgrazing of some areas
 - ✓ Tilling along river banks

Importance of Responsible Utilization of the Resources

- ✓ Minimize pollution, habitat destruction, and biodiversity loss.
 - ✓ Ensure resources are available for future generations.
 - ✓ Avoid resource scarcity that could drive up costs or create conflicts.
 - ✓ Reduce greenhouse gas emissions by prioritizing renewable over non-renewable resources.
- ❑ By utilizing resources wisely and balancing consumption with conservation, society can meet its current needs while protecting the planet for the future.

Teaching and Learning Materials

- ❑ Posters and Infographics
- ❑ Videos and Documentaries

- ☐ Interactive Maps
- ☐ Hands-on Kits
- ☐ Textbooks and Articles
- ☐ Guest Lectures

Expected Standard

- ☐ Renewable and non-renewable resources used responsibly.

**Assessment**

- ☐ Learners to carry out a simple research on the use of renewable and non-renewable natural resources responsibly.
- ☐ Practice clean technologies, conservation and restoration methodologies in utilizing natural resources responsibly.
- ☐ Learners to conduct debate or quiz on utilisation of renewable and non-renewable resources

1.1.2. SUB-TOPIC 2: POPULATION AND NATURAL RESOURCES**Introduction**

The relationship between population and natural resources is a critical element in understanding the dynamics of human development and environmental sustainability. As the Zambian population continues to grow, the demand for essential resources such as water, land, energy, and minerals increases exponentially. Natural resources are the foundation of human survival and economic activity, providing the raw materials for food, shelter, industry, and energy. However, these resources are finite, and their overexploitation can lead to environmental degradation, scarcity, and conflict.

Population growth amplifies pressure on ecosystems, often leading to unsustainable practices in agriculture, mining, deforestation, and energy consumption. In densely populated regions, the strain on natural resources can outpace the Earth's capacity to regenerate them, resulting in pollution, habitat loss, and a decline in biodiversity.

Specific competence – Learners to:

Analyze the relationship between population and natural resources.

**Activity 1**

Discussing Population and Natural Resources

Learning Environment

The learning environment would be outdoor or indoor set up.

Indoor Set up:

- ☐ Briefly introduce the relationship between population and natural resources, emphasising concepts such as resource scarcity, sustainability, and overutilization.
- ☐ Pose a question: *“What happens when population growth exceeds resource availability?”*
- ☐ Assign each group a specific aspect to discuss (such as water use, energy consumption, deforestation, or agricultural land).
- ☐ Groups will analyse how population affects resource availability in their assigned aspect.

- ☐ Learners use sticky notes or charts to record their findings, identifying both challenges and opportunities.
- ☐ Groups present their findings to the class.
- ☐ Encourage peer questions and help refine ideas further.

Outdoor Set up:**Preparation**

- ☐ Select an outdoor location where learners can observe natural surroundings, human activity, and environmental impacts (e.g., school grounds, nearby park, or community area)
- ☐ Divide learners into small groups of 4–6 members.
- ☐ Briefly explain the activity, focusing on how population growth and human activities can impact the environment (e.g., deforestation, pollution, overuse of resources).
- ☐ Provide examples of what to look for, such as litter, damaged vegetation, or evidence of resource use.

Field Observation

- ☐ Each group explores the outdoor area and identifies visible effects of population on the environment.
- ☐ Encourage learners to take notes and make sketches or diagrams to document what they observe.
- ☐ Groups can categorize their findings, such as “pollution,” “loss of vegetation,” or “resource overuse.”

Discussion

- ☐ Groups reconvene to discuss their findings.
- ☐ Learners analyse the causes behind the observed environmental impacts and consider how population activities might contribute (e.g., “We saw litter near the park, which might be due to an increase in visitors without proper waste disposal.”).

**Content Tips**

- ☐ The Link between population and resource demand
 - ✓ Population increase results in high demand for basic necessities such as food, water, energy, and living space.
 - ✓ Larger populations require more infrastructure, transportation, housing, and services, leading to pressure on natural resources.
 - ✓ Rising standards of living (especially in developing nations) increase per capita resource consumption.
- ☐ Overconsumption outpaces natural replenishment, leading to:
 - ✓ Soil degradation
 - ✓ Water shortages
 - ✓ Loss of biodiversity
 - ✓ Air and water pollution
- ☐ Environmental degradation from overpopulation can result in:
 - ✓ Over-cultivation and desertification
 - ✓ Overfishing and ocean depletion
 - ✓ Excessive carbon emissions and climate change
- ☐ Lack of balance in the ecosystem are created when human activity outstrips the environment’s capacity to regenerate.

- ❑ Impacts of overpopulation could be addressed through technological advancements such as:
 - ✓ Renewable energy (solar, wind)
 - ✓ Water purification and recycling
 - ✓ Climate-SMART agriculture practices such as crop rotation, agroforestry, mulching etc and sustainable agriculture methods.
- ❑ To manage the population-resource relationship;
 - ✓ Promote education and family planning to slow population growth.
 - ✓ Encourage sustainable development and green technologies.
 - ✓ Implement environmental laws and international agreements (e.g., Paris Agreement).
 - ✓ Shift to circular economies where resources are reused and recycled.

Teaching and Learning Materials

- ❑ Posters and Infographics on The 3Rs (Reduce, Reuse, Recycle)
- ❑ Charts and Diagrams showing population and natural resources
- ❑ Videos and Documentaries on population and natural resources



Assessment

Ask learners to:

- ✓ suggest ways on how to reduce the pressure of population on natural resources.
- ✓ debate different viewpoints on managing population and natural resources.
- ✓ role play: “A village meeting discussing a water shortage due to increased population.”
- ✓ debate on “Technology vs. Population Control-Which is the better solution for managing resources?”



Activity 2

Analysing the relationship between population and natural resources

Learning Environment

The learning activity would be indoor or outdoor set up.

Indoor set up:

- ❑ Learners in groups discuss how people relate with the natural resources.
- ❑ Discuss the impact of large population on the natural resources

Outdoor set up:

- ❑ Learners to visit the local environment and observe how people are relating with the natural resources.
- ❑ Learners assess some human activities that may be a threat to the natural resources.



Content Tips

- ❑ The relationship between population and natural resources is dynamic and complex, as population growth and resource consumption are interdependent.
- ❑ As the population grows, the demand for natural resources increases. This includes resources for:

- **Food:** The demand for agricultural land, water and raising livestock increases while the land is limited.
 - **Energy:** The demand for energy such as biomass, hydroelectricity for heating and transportation leads to eventual issues such as pollution.
 - **Water:** The demand for freshwater for domestic use and irrigation accelerates as the population increases.
 - **Raw materials:** There is also high demand for minerals, metals, and timber for construction, manufacturing, and infrastructure.
- ☐ As cities expand, there is an increase in resource demand for construction, water and energy supply.
 - ☐ As populations exceed carrying capacities, competition for resources intensifies, potentially leading to conflicts, higher costs, and social inequality.
 - ☐ Population growth accelerates depletion and creating scarcity of natural resources.
 - ☐ The relationship between population and natural resources is reciprocal. A growing population increases resource demand, while resource availability influences population well-being and growth patterns.
 - ☐ Sustainable practices, technological innovation, and equitable resource distribution are critical to maintaining this balance and ensuring the well-being of both people and the planet.

Teaching and Learning Materials

- ☐ Charts and Graphs illustrating population growth trends, resource consumption and deforestation rates.
- ☐ Documentaries and videos showing real-life examples of resource depletion and conservation efforts in Zambia.
- ☐ Posters displaying information on sustainable resource use and conservation.
- ☐ Online Resources: Use websites and digital platforms to access data on Zambia's population and natural resources.
- ☐ Mobile Applications: Interactive applications for environmental education and sustainable resource management.



Assessment

- ☐ Learners to create posters or presentations advocating for sustainable resource management.
- ☐ Learners propose community-based solutions that can help balance population growth with resource conservation.
- ☐ Learners to critically analyze and discuss issues to do with the relationship between population and natural resources.
- ☐ Learners to debate on the impact of overpopulation on a given natural resource in Zambia.



Activity 3

Engage in practices that ensure *(efficient use of resources, conservation of water resources, use of renewable energy, reducing waste, sustainable consumption, advocating for family planning...)*

Learning Environment

The learning environment would be indoor or outdoor set up.

Indoor set up:

- ☐ Learners in various groups to discuss strategies that ensure efficient use of natural resources.
- ☐ Each group to share key ideas to the class.
- ☐ Key responses from each group to be written down by grouping similar ideas.

Outdoor set up:

- ☐ Learners to conduct a nature walk to observe and appreciate natural resources around them
- ☐ Learners to participate in recycling activities by collecting recyclable materials to promote waste reduction.

**Content Tips**

- ☐ Efficient use of natural resources refers to optimal utilization of natural resources such as water, energy, land and minerals while minimizing waste and environmental degradation
- ☐ Promote natural resources management practices such as Reduce, Reuse and Recycle.
- ☐ Promote energy efficiency ways to reduce energy consumption such as use of efficient appliances and turning off lights when not in use.
- ☐ Water conservation to reduce water waste
- ☐ Sustainable land use practices like fallowing, crop rotation, agroforestry etc
- ☐ Avoid cutting trees anyhow to reduce deforestation

Teaching and Learning Materials

- ☐ Charts or pictures on natural resource management
- ☐ Videos on natural resource conservation
- ☐ Journals on environmental science and conservation of resources

Expected Standards

- ☐ Relationship between Population and Natural resources analysed correctly.

**Assessment**

- ☐ Assess learners as they design a project for reducing waste and conserving natural resources in a school or community.
- ☐ Observe learners as they engage in ways of conserving water like fixing leakages and turning off taps when not in use.

1.1.3. SUB-TOPIC 3: SUSTAINABLE MANAGEMENT OF NATURAL RESOURCE.**Introduction:**

Sustainable management means using natural resources in a careful and responsible way so that they last for a long time. Sustainable management of natural resources aims to balance human needs with environmental protection, ensuring that these resources are available for current and future generations. This approach promotes responsible use, conservation, and restoration strategies that align with ecological principles and long-term sustainability goals. By integrating environmental, social, and economic considerations, sustainable resource management plays a crucial role in building a more resilient and equitable world.

Specific Competence(s) – Learners to;

Practise sustainable management of natural resources



Activity 1

Identifying sustainable ways of managing natural resources

Learning Environment

The learning environment would be indoor or outdoor set up.

Indoor set up:

- ☐ Learners work in small groups to identify sustainable ways of managing natural resources.
- ☐ Each group presents their findings to the class.

Outdoor set up:

- ☐ Learners go around the school compound (or designated area) in small groups, identifying natural resources they see (trees, soil, water tanks, grass, stones, etc.).
- ☐ Learners take notes or sketch drawing of each resource.
- ☐ For each resource identified, learners to discuss how it's used and how it can be cared for.
- ☐ Learners participate in a short clean-up activity in the school surrounding to practice sustainable environmental care.



Content Tips

- ☐ Sustainable Ways of Managing Natural Resources include;
 - ✓ Water: use clean water wisely, fix leaking taps, harvest rainwater
 - ✓ Soil: plant trees, crop rotation, use manure instead of chemicals
 - ✓ Forests: plant more trees (reforestation), avoid illegal logging
 - ✓ Air: avoid burning plastics, use clean energy (e.g., solar)
 - ✓ Minerals: recycle, mine responsibly, reuse materials
- ☐ Roles of Individuals and Communities
 - ✓ Educating others about conservation
 - ✓ Participating in environmental clean-up and tree planting
 - ✓ Reporting harmful activities (like illegal logging or pollution)
 - ✓ Practicing recycling and proper waste disposal
 - ✓ Emphasize the idea of “*using without wasting*” as a simple way to explain sustainability.

Teaching and Learning Materials

- ☐ Charts with information on sustainable ways of managing natural resources
- ☐ Sketch books or writing pads
- ☐ Sample pictures or flashcards of natural resources
- ☐ Waste collection bags/gloves (for clean-up activity)

Assessment

- ☐ Observe learners during group work and outdoor activities.
- ☐ Ask learners oral questions during and after activities.

- ☐ Ask learners to explain some sustainable ways of managing the natural resources.
- ☐ Ask learners to make models using collected waste materials.



Activity 2

Practising sustainable management of natural resources.

Learning Environment

The learning activity would be indoor or outdoor set up.

Indoor set up:

- ☐ Learners in groups suggest practices that promote sustainable resource management.
- ☐ Learners move around the school and local environment and identify practices that promote sustainable resource management.

Outdoor set up:

- ☐ Learners to tour a local conservation area or eco-farm to observe various conservation methods.
- ☐ Learners to visit a water treatment or recycling plant.
- ☐ Learners to tour the local environment and identify various practices on sustainable resource management.



Content Tips

- ☐ Sustainable management of natural resources involves careful and efficient use of resources in a manner that ensures their availability for future generations while balancing ecological, economic, and social needs.
- ☐ It focuses on reducing resource depletion, minimizing environmental degradation, and ensuring that ecosystems remain resilient to changes such as population growth and climate change.
- ☐ Sustainable management is not only about conserving resources but also about using them in ways that support long-term prosperity and equity.
- ☐ Key practices for the sustainable management of natural resources include;
 - ✓ Sustainable Water Management through Efficient Irrigation, i.e. drip and sprinkler systems
 - ✓ Rainwater Harvesting
 - ✓ Water Recycling and Treatment
- ☐ Forest and Land Management through;
 - ✓ Sustainable Logging i.e. selective logging
 - ✓ Agroforestry
 - ✓ Land Restoration i.e. Rehabilitating degraded land
 - ✓ Controlled Grazing, i.e. rotational grazing
- ☐ Sustainable Agriculture through;
 - ✓ Crop Rotation
 - ✓ Organic Farming
 - ✓ Conservation Tillage
 - ✓ Diversified Farming
- ☐ Energy Efficiency, such as LED lighting
- ☐ Recycling and Reusing

- ☐ Composting i.e use of organic wastes
- ☐ Community Engagement and Education
 - ✓ Public Awareness Campaigns
 - ✓ Community-Based Resource Management

Teaching Materials

- ☐ Posters and Infographics on The 3Rs (Reduce, Reuse, Recycle)
- ☐ Charts and diagrams showing sustainable management practices of natural resources.
- ☐ Videos and documentaries depicting practices on sustainable management of natural resources.

Expected Standard

Sustainable management of natural resources practised appropriately



Assessment

- ☐ Learners to create a sustainability action plan for the school and implement it.
- ☐ Learners role-play as policymakers in creating sustainability laws.
- ☐ Learners perform composting activity to promote organic waste management.



Summary

Renewable and non-renewable natural resources

- ☐ To ensure a sustainable future, it is important to use both types of resources responsibly. This means:
- ☐ Conserving non-renewable resources by reducing waste, using energy-efficient technologies, and switching to alternative energy sources.
- ☐ Managing renewable resources by avoiding overexploitation and promoting practices like reforestation and sustainable farming.
- ☐ Encouraging recycling and reuse to minimize resource depletion and environmental harm.
- ☐ Raising awareness and educating communities about the importance of sustainability.
- ☐ Responsible resource use helps protect the environment, supports economic stability, and ensures that future generations can also meet their needs.

Relationship between population and natural resources

- ☐ There is a strong connection between population growth and the use of natural resources.
- ☐ As the population increases, the demand for resources such as water, food, energy, land, and raw materials also rises.
- ☐ A larger population leads to:
 - ✓ Greater consumption of both renewable and non-renewable resources.
 - ✓ Increased pressure on the environment through deforestation, pollution, and overfishing.
 - ✓ Resource depletion, especially when consumption is faster than the natural replenishment rate.
- ☐ However, the impact of population on resources also depends on factors like technology, lifestyle, and resource management. For example:
 - ✓ Countries with high consumption rates per person can have a larger environmental impact than more populous but lower-consuming nations.

- ❑ Efficient technologies and sustainable practices can reduce resource strain.
- ❑ Understanding this relationship is important for promoting sustainable development, where the needs of the present are met without compromising the ability of future generations to meet theirs.

Sustainable management of natural resources

- ❑ Sustainable management of natural resources involves using Earth's resources—such as water, soil, forests, minerals, and fossil fuels—in ways that meet current human needs without compromising the ability of future generations to meet theirs.
- ❑ It emphasizes a balance between environmental protection, economic development, and social well-being.
- ❑ Key practices include:
 - ✓ Conservation: Protecting ecosystems and biodiversity through measures like protected areas, reforestation, and wildlife preservation.
 - ✓ Efficient resource use: Reducing waste and maximizing the efficiency of water, energy, and raw materials.
 - ✓ Renewable energy: Shifting from fossil fuels to renewable sources like solar, wind, and hydro power.
 - ✓ Community involvement: Engaging local communities in decision-making and promoting traditional knowledge and practices.
 - ✓ Policy and regulation: Enforcing laws that regulate extraction, pollution, and land use to prevent overexploitation and degradation.
 - ✓ By adopting these strategies, societies can ensure long-term resource availability, reduce environmental impacts, and promote resilience against climate change and other global challenges.

2.1. TOPIC 2: POPULATION

Introduction

Population studies also known as demography, is the scientific investigation of human populations, focusing on their size, structure, distribution and dynamics. This topic is important in understanding the complexities of human populations and their impact on the environment, economy and society. Understanding population dynamics help education policies and policy makers to address challenges of infrastructure, youth unemployment, skill development, entrepreneurship, environmental degradation and service delivery. The focus of this topic is on;

- ☐ Population composition
- ☐ Distribution
- ☐ Rapid population growth

Hook: *Imagine you were the mayor of a fast-growing city; how would you plan for the increasing population?*

General Competences

The general competencies in this topic are: analytical thinking, creativity and innovation, digital and problem solving.



Key Terms/Words/Vocabulary

- ☐ Population
- ☐ Demography
- ☐ Census
- ☐ Population density
- ☐ Birth rate
- ☐ Death rate
- ☐ Migration

2.1.1. SUB –TOPIC 1: POPULATION COMPOSITION

Introduction

Population composition refers to the various characteristics that make a population, such as age, gender, density and ethnicity. The other elements of population composition include distribution and density, wealth, education, marital status, fertility, and mortality. Population structure (also referred to as population composition) is the distribution of different social groups across a nation or area. There are many various aspects to learn about when studying population structure, such as: the meaning of population structure, the characteristics of the structure, the factors impacting these characteristics, how to measure it, and the importance of the structure for a country. Population structure is the breakdown of different groups of people in an area. It is important as it affects the area itself. Population pyramids vary in shape from country to country, depending on their level of economic development.

Specific Competence(s) – Learners to:

Demonstrate understanding of population composition.



Activity 1

Exploring population concepts.

Learning Environment

This learning environment can be done in doors

Indoor set up:

- ☐ Divide learners into groups. Assign each group a specific population concept to focus on (such as population growth, distribution patterns, or population density).
- ☐ Encourage each group to brainstorm and identify real-world examples related to their assigned concept.
- ☐ Groups discuss their assigned concept, identifying key ideas, trends, and implications (e.g., how population growth affects resources).
- ☐ Groups prepare a visual representation or chart summarizing their findings.



Content Tips

- ☐ **Population Density:** It is the number of people living per square kilometer in an area. Population density may be high or low.
- ☐ **Population Distribution:** It is the way the population is spread in a given area or space. Some places are sparsely, moderately or densely populated.
- ☐ **Population Structure:** It is made up of different age groups and sex. Population structure is shown using population pyramids.
- ☐ **Birth Rate:** It is the number of babies born live in every 1000 people in a population in a year.
- ☐ **Crude Birth rate:** is the total number of babies born per 1000 people in a population within a given period of time expressed as a percentage. Example: if 40 births occur, this means $40/1000 \times 100 = 4\%$.
- ☐ **Death Rate:** It is the number of deaths happening in every 1000 people of a population.
- ☐ **Crude Death rate;** It is the number of death per 1000 people in an area over a given period of time expressed as a percentage. This is also known as mortality rate. Example: if 50 deaths occurred, this means $50/1000 \times 100 = 5\%$.
- ☐ **Migration:** This is the movement of people from one place to another for settlement. It could be within the country (internal) or between countries (international). Migration can either lead to a decrease or increase in population of an area or country.

Teaching and Learning Materials

- ☐ Charts showing calculation of population concepts
- ☐ Pictures or charts of population pyramids



Assessment

- ☐ Observe learners' participation in small groups as they discuss population concepts.
- ☐ Check the correctness of the learners' calculations related to population concepts.
- ☐ Monitor the responses of learners as they attempt quiz questions on population concepts.



Activity 2

Drawing population pyramids to show population composition (age structure, sex ratio...)

Learning Environment

The environment set up for this activity can be both indoor and outdoor:

Indoor set up:

- ☐ Arrange desks in small groups of 4–6 learners to promote teamwork.
- ☐ Provide each group with printed copies or digital displays of various population pyramids. Use examples that differ in structure and composition (e.g., broad base, narrow base, gender balance).
- ☐ Briefly explain the purpose of population pyramids and what they represent. Highlight the key elements (age groups, gender distribution, and structure).
- ☐ Show an example pyramid and demonstrate how to interpret it, noting its trends and patterns (e.g., high birth rates, aging population, etc.).

Each group will:

- ☐ Analyse the provided population pyramids by identifying differences and similarities in their structures (e.g., broad vs. narrow base).
- ☐ Use note books or charts to write down observations and insights (e.g., “This pyramid shows a high dependency ratio because...”).
- ☐ Discuss factors that might lead to these structures (e.g., high fertility, urbanization, migration).

Presentation

- ☐ Groups share their analyses and solutions with the class.
- ☐ Encourage other groups to ask questions or add insights.

Outdoor set up:**Preparation**

- ☐ Form small groups of 4–6 learners to foster collaboration.
- ☐ Brief learners on population pyramids, explaining their components (age groups, gender distribution) and how they reflect socioeconomic characteristics

Population Pyramid Drawing

- ☐ Groups will draw different population pyramids on the ground or large paper, representing less income countries, medium-income countries, and high-income countries.
- ☐ Use materials (e.g., colored chalks or markers) to differentiate age groups and genders.
- ☐ Learners analyse and compare the population pyramids for different countries discussing the implications of the population composition.

**Content Tips**

- ☐ Population pyramids is a graph that shows the number of people in different age and sex groups in a population.
- ☐ Population pyramids vary in shape from country to country, depending on their level of economic development.
- ☐ Population pyramids allow population structures to be presented in a way that makes the data easy to analyse and compare.
- ☐ They can be used to compare changes in the structure of the population over time.
- ☐ Population pyramids in Low-Income Countries (LIC) typically have a wide base and a narrow top.

- ☐ The top of the population pyramid is also wider than a LIC because people are living longer due to improvements in health care creating an ageing population.
- ☐ Population pyramids in High Income-Countries (HICs) are very different to LICs. They typically have a narrower base, indicating lower birth rates. There is also a wide central section showing a large working population.
- ☐ Population pyramids may contain significant anomaly in terms of the considerable bulge on the side of males between 20 and 59 years for instance towns with mining and industrial activities e. g Solwezi.

Teaching and Learning Materials

Charts showing population pyramids of different countries in different categories



Assessment

- ☐ Ask learners to explain anomalies in population pyramids
- ☐ Observe learners as they draw different population pyramids representing a particular grouping (e. g, Low Income-Countries, Middle Income-Countries and High Income-Countries).



Activity 3

Interpreting population composition using population pyramids (*age distribution and its implications on resources/needs of different age groups in the community...*)

Learning Environment

The environment set up for this activity can be both indoor and outdoor:

Indoor Set up:

- ☐ In groups learners discuss and interpret population composition using population pyramids by identifying trends, patterns and correlations.
- ☐ Discuss the implication of population composition on resources or needs of different age group in the community.
- ☐ Using online tools or GIS software to create interactive population pyramid models

Outdoor Set up:

- ☐ Divide learners into age groups; organize learners into different assumed age groups (e. g. 0 -10, 11-14, 15-18, 19-24) to form population pyramid.
- ☐ Form population pyramid; ask each assumed age group to stand in a line, with the youngest age group at the base and the oldest at the top.
- ☐ Learners to analyse and discuss the shape of the population pyramid, exploring how it reflects the population composition, potential implications and needs.



Content Tips

- ☐ Population pyramids can be split in half, showing males on the left and females on the right.
- ☐ The vertical axis is usually divided into 5-year intervals.
- ☐ The values along the x-axis can be raw numbers but are usually expressed as a percentage of the population.
- ☐ Population pyramids emphasise on population growth, dependency ratio and age structure.

- ☐ Population pyramids help in identifying resource use in the area e. g age group between 20 and 50 years tend to use more resources.
- ☐ Youthful population will have high demand on education and child care services, increased pressure on job market while an aging population will put increased demand for health care and social services.

Teaching and Learning Materials

- ☐ Maps showing population pyramids
- ☐ Diagrams, Charts and Graphs depicting different population pyramids.
- ☐ Videos or documentaries on population pyramids

Expected Standard

Understanding of population composition demonstrated accordingly

**Assessment**

- ☐ Observe learners as they;
 - ✓ discuss the impact of an aging population on health care and social security.
 - ✓ identify patterns and trends in population composition across different regions.
 - ✓ discuss factors that influence population composition such climate, terrain and natural resources.

2.1.2. SUB-TOPIC 2: POPULATION DISTRIBUTION IN ZAMBIA**Introduction**

Zambia, a land locked country in Southern Africa, has a diverse and complex population distribution. With a total population of 19.6 million people as at 2022 Census. Zambia's population is unevenly distributed across its 10 Provinces. The country's population distribution is influenced by various factors, including geography, climate, economy and history. Understanding population distribution is crucial for effective planning, resource allocation and decision making in various sectors, such as healthcare, education and infrastructure development. This topic will explore the patterns, trends and factors influencing population distribution in Zambia.

Specific Competence(s) – Learners to:

Analyze population distribution

**Activity 1**

Locating on the Map of Zambia areas of sparse, moderate and dense population.

Learning Environment

The environment set up for this activity can be both indoor and outdoor

Indoor Set up:

- ☐ Learners in small groups to participate in the discussions.
- ☐ Provide maps on population distribution patterns in Zambia.
- ☐ Learners identify sparsely, moderately and highly populated areas in Zambia.

Outdoor Set up:

- ☐ Learners to create maps showing population distribution patterns in a selected area.
- ☐ Learners to analyse maps showing population distribution patterns in a selected area.
- ☐ Learners conduct surveys in local communities to gather data on population distribution and factors influencing it.



Content Tips

Areas of Low Population Density

- ☐ These comprise of mountainous areas such as Mafinga, Makutu and Mbala, Muchinga Escarpment, Zambezi Escarpment and areas of low rainfall such as Luangwa and Zambezi Valleys, most parts of Western and Northwestern Provinces.
- ☐ Other areas include Mweru-Wantipa, Game parks, Game and Forest reserves.

Areas of Medium population Density

- ☐ The medium populated areas are those that receive adequate rainfall and mostly are rural areas in which farming is the main activity.
- ☐ These areas include rural provinces of Eastern, Southern, Central, Luapula, Northern and Muchinga.

Areas of High population Density

- ☐ These are areas of large population comprising of urban areas especially Provincial capitals.
- ☐ Much of Zambia's population is concentrated around Lusaka and Copperbelt because of many industries and mining activities.

Teaching and learning Materials

- ☐ Maps and Atlases and journals showing population distribution in Zambia.
- ☐ Case studies and Reports from Zambia Statistical Agency on population distribution in Zambia.



Assessment

- ☐ Check learners as they analyze and interpret maps showing population distribution patterns in Zambia.
- ☐ Learners participate in quizzes to assess their understanding of population distribution



Activity 2

Describing factors influencing population distribution in Zambia (*water, electricity, minerals, transport...*)

Learning Environment

The environment set up for this activity can be both indoor and outdoor

Indoor Set up:

- ☐ Divide learners into small groups of 4–6 members to encourage teamwork and interaction.
- ☐ Provide a brief explanation of population distribution and its importance. Use visual aids, like a simple map of Zambia to highlight varying population densities.
- ☐ Each group will select one factor (e.g., water availability, altitude, soil fertility, transportation, minerals, or electricity).
- ☐ Discuss how that factor influences population distribution in a specific area or region.
- ☐ Prepare a brief presentation or visual representation of their findings (e.g., a mind

map, poster, or chart)

Outdoor Set up:**Preparation Phase**

- ☐ Provide an overview of settlement factors (such as availability of water, fertile soil, transportation networks, resources, etc.).
- ☐ Equip learners with basic survey techniques (how to design questions, conduct interviews, and record responses).

Field Survey in the Local Community

- ☐ Form groups of 4–6 members to encourage collaboration. Assign each group a specific focus area
- ☐ Each group will visit a designated area in their local community.
- ☐ Conduct surveys by interviewing community members to understand why people settled in that location and what factors continue to shape the area's development.
- ☐ Use observation skills to identify visible factors (such as proximity to roads, availability of utilities like electricity or water).

Post Activity Presentation

- ☐ Back in class, each group will analyse their findings and prepare a brief presentation (using charts, posters, or short slideshows...).
- ☐ Groups will present their work to their classmates, encouraging peer feedback and questions.

**Content Tips****Factors Affecting Population Distribution****Natural Factors**

- ☐ Climate: Human habitation is favored by areas that receive adequate rainfall and moderate temperatures.
- ☐ Relief: most densely populated areas are flat land with fertile soils.
- ☐ Availability of water: plenty of water attracts settlements and have high potential for economic development.
- ☐ Fertile soils: attract people due to farming opportunities.
- ☐ Availability of natural resources: many people settle in places where there is enough resources to exploit.

Social Factors

- ☐ Most people tend to settle in areas where many social amenities are available such as good health facilities, education, housing, social clubs, malls and social security.

Economic Factors

- ☐ Economic factors such as business opportunities, employment, transport and communications services, electricity, shopping malls, availability of goods, industries and factories attract settlements.

Political Factors

- ☐ Government resettlement programs conducted under local government settlement plans and colonial influences determine location of settlements.

Teaching and Learning materials

- ☐ Maps highlighting population distribution patterns in Zambia.
- ☐ Case Studies of real- life examples of how different factors affect population distribution in different regions.
- ☐ Graphs and Charts illustrating the relationship between factors affecting location of settlements and population distribution.



Assessment

- ☐ Ask the learners to;
 - ✓ identify and explain population distribution patterns on a map.
 - ✓ discuss the factors influencing population distribution.
 - ✓ explain how water, electricity, transport and minerals influence population distribution.



Activity 3

Analysing population distribution using Maps or Geographical Information Systems (GIS)

Learning Environment

The learning environment would be both indoor and outdoor set up

Indoor Set up:

- ☐ Divide learners into groups.
- ☐ Provide maps or use projector or smart board to display maps and GIS data on population distribution.
- ☐ Provide access to computers with GIS software or online mapping tools.

Outdoor Set up:

- ☐ Conduct field trip within a nearby area to collect data on population distribution using GPS devices or mobile phones app.
- ☐ Set up an outdoor laboratory with GIS equipment and software for learners to collect and analyse data.



Content Tips

- ☐ Maps and Geographic Information Systems (GIS) are essential tools for analysing and understanding population distribution. They enable users to visualize, analyse and interpret population data in relation to geographic locations
- ☐ Maps provide a visual representation of population distribution patterns, making it easier to identify trends and patterns.
- ☐ Maps enable spatial analysis in examining the relationships between population distribution and geographical factors like climate, topography and infrastructure.
- ☐ Maps can integrate multiple data sources, such as census data, satellite imagery and GPS data, to provide a comprehensive understanding of population distribution.
- ☐ GIS software provides accurate and reliable data and advanced data analysis capabilities, enabling users to analyse population distribution data in relation to geographical locations.
- ☐ GIS allows users to create spatial models that predict population distribution patterns based on various factors like climate change, economy and infrastructure.

- ❑ GIS provides decision makers with accurate and reliable data to inform policy decisions relating to population distribution, urban planning and resource allocation.

Teaching and Learning Materials

- ❑ Maps, Mobile Apps, GIS software, GPS device, Open Street Map, Google Earth, Google maps.

Expected Standard

- ❑ Population distribution analyzed appropriately.

**Assessment**

- ❑ Ask learners to;
 - ✓ analyse population distribution patterns using maps or GIS.
 - ✓ identify factors influencing population distribution.
 - ✓ explain the implication of population distribution patterns on economic, social and environmental development.

2.1.3. SUB – TOPIC 3: RAPID POPULATION GROWTH IN ZAMBIA**Introduction**

Zambia has experienced significant population growth over the past few decades, driven by factors such as high birth rates, declining death rates, and immigration. This rapid increase in population presents both opportunities and challenges, including pressure on social services, infrastructure, and natural resources. Understanding the causes of this growth is essential for planning sustainable development and ensuring a better quality of life for future generations.

Specific Competence(s) – Learners to:
Address challenges of rapid population.

**Activity 1**

Analysing the Causes of Rapid Population Growth in Zambia (high birth rate, declining death rate, early marriage, immigration, illiteracy...)

Learning Environment

The learning environment would be both indoor and outdoor set up

Indoor Set up:

- ✓ Learners in groups to discuss the causes of rapid population growth.
- ✓ Group presentations on each contributing factor (e.g., high birth rate, early marriages, illiteracy).
- ✓ Watching videos or presentations on Zambia's demographic trends by Zambia Statistics Agency.
- ✓ Case study analysis of population growth in different regions of Zambia.

Outdoor Set up:

- ❑ Field trips to local health centers to understand demographic trends.
- ❑ Surveys and interviews with community members about factors influencing family size.



Content Tips

Rapid population growth is caused by a number of factors such as;

- ☐ High Birth Rates: Zambia has a high Total Fertility Rate (TFR) of 4.7 children per woman as 2022 census, contributing to rapid population growth.
- ☐ Low Birth Rates: The number of people dying is much lower than those that are born
- ☐ Early Marriages: Early marriages and childbearing are common in Zambia, leading to a higher number of births.
- ☐ Limited Access to Family Planning: Many Zambians, particularly in rural areas, lack access to family planning resources and education.
- ☐ Low Literacy Rates: Low literacy rates, especially among women, contribute to a lack of understanding about family planning and reproductive health.
- ☐ Limited Access to Schools: Many Zambian children, particularly girls, do not have access to education, leading to earlier marriages and childbearing.
- ☐ Large Family Sizes: In some Zambian cultures, large family sizes are valued, leading to a desire for many children.
- ☐ Traditional Beliefs: Some traditional beliefs and practices, such as the importance of having many children for labor and old-age security, contribute to high birth rates.
- ☐ Poverty: Poverty is a significant driver of population growth in Zambia, as families may rely on children for labor and economic support.
- ☐ People coming to settle in Zambia from other countries

Teaching and Learning Materials

- ☐ Charts and graphs showing Zambia's population trends.
- ☐ Videos or documentaries on population growth.
- ☐ Articles on demographic changes.
- ☐ Interview guides for interviews and surveys.



Assessment

- ☐ Conduct Question-and-answer sessions during discussions on rapid population growth.
- ☐ Facilitate class debates on the advantages and disadvantages of rapid population growth.
- ☐ Assess written essays on the causes and effects of population growth.
- ☐ Assign project-based assessments such as a population growth report or poster presentation.



Activity 2

Analysing the effects of Rapid Population Growth in Zambia (unemployment, high crime rates, pressure on social services, overcrowding...)

Learning Environment

The learning environment would be both indoor and outdoor set up

Indoor Set up:

- ☐ Learners in groups discuss how population growth affect employment, crime, and social services.
- ☐ Learners to debate on whether population growth is more beneficial or harmful.

- ☐ Learners to watch documentaries or news reports on urban overcrowding and unemployment in Zambia.
- ☐ Learners to analyse data using charts and graphs showing trends in crime, unemployment, and public service demands.

Outdoor Set up:

- ☐ Field visits to overcrowded communities to observe the impact on services delivery.
- ☐ Surveys or interviews with community members about how population growth affects their daily lives.
- ☐ Role-playing exercises where students act out real-life scenarios, such as job-seeking.
- ☐ Observing and documenting overcrowded public transport or markets.



Content Tips

- ☐ Rapid population growth has a number of effects. Some of them include:
- ☐ Unemployment: As the population grows, job opportunities become scarce, leading to high unemployment rates, especially among the youth.
- ☐ Crime Rate: High unemployment and poverty can drive people into criminal activities such as theft and drug abuse.
- ☐ Pressure on Social Services: Schools, hospitals, and housing become overcrowded, reducing the quality of services provided.
- ☐ Overcrowding: Cities become densely populated, leading to poor living conditions, inadequate sanitation, and traffic congestion.
- ☐ Encourage students to think critically about possible solutions and policies to manage these challenges.

Teaching and Learning Materials

- ☐ Charts and graphs showing Zambia's unemployment and crime trends.
- ☐ Videos/documentaries on urban overcrowding.
- ☐ Articles and government reports on the effects of rapid population growth.
- ☐ Survey forms for fieldwork.
- ☐ Role-play scripts for interactive learning.



Assessment

- ☐ Coordinate class discussions and reflections on observed field experiences.
- ☐ Facilitate group presentations on specific effects of population growth.
- ☐ Conduct quick quizzes on key concepts discussed in class.
- ☐ Assign group project to write a proposal on how to address specific population-related issue in Zambia.



Activity 3

Describing the Potential Benefits of Rapid Population Growth in Zambia (adequate manpower, maximum utilisation of natural resources, security...)

Learning Environment

The learning environment would be both indoor and outdoor set up

Indoor Set up:

- ☐ Learners in groups to discuss how population growth can boost the economy.
- ☐ A documentary or a Case study of countries that have benefited from population growth.
- ☐ Debates on whether Zambia's population growth is more beneficial or problematic.

Outdoor Set up:

- ☐ Field visits to industries or farms to observe how population growth supports labor supply.
- ☐ Interviews with business owners on how population growth affects the workforce and promote businesses.
- ☐ Surveys on community perspectives regarding population benefits.

**Content Tips**

Rapid population growth has a number of potential benefits such as:

- ☐ Adequate Manpower: A larger population provides a workforce for industries, agriculture, and services, boosting economic growth.
- ☐ Maximum Utilization of Natural Resources: More people mean greater capacity to explore and utilize Zambia's abundant resources, such as minerals, fertile land, and water bodies.
- ☐ Security and National Defense: A larger population strengthens national security by increasing the number of available personnel for defense and law enforcement.
- ☐ Market Expansion: A growing population increases demand for goods and services, encouraging entrepreneurship and business growth.
- ☐ Expanded tax base: a large population means a broader tax base providing more revenue for the Government.
- ☐ Increased innovation: a larger population can lead to more innovations as people from different backgrounds and experiences come together to share ideas and solve problems.

Teaching and Learning Materials

- ☐ Charts and graphs on population growth and economic impact.
- ☐ Videos or case studies of successful economies with high population growth.
- ☐ Articles and reports on Zambia's labor force and resource utilization.
- ☐ Interview guides for surveys and interviews.
- ☐ Maps of Zambia showing resource distribution and economic activities.

**Assessment**

- ☐ Facilitate group discussions and reflections on field experiences.
- ☐ Administer essays on how Zambia can maximize the benefits of population growth.
- ☐ Assign group project proposing strategies for sustainable population growth management.

**Activity 4**

Engaging in Actions that help Address the Challenges of Rapid Population Growth (improving waste management resource management, raising awareness...)

Learning Environment

The learning environment will be both indoor and outdoor set up

Indoor Set up:

- ☐ Learners in groups to discuss strategies on how to manage population growth challenges.
- ☐ In groups, learners brainstorm on innovative solutions for waste management and resource conservation.
- ☐ Learners to watch educational videos on sustainable living and environmental conservation.
- ☐ Learners to debates on government policies and community initiatives to manage rapid population growth.

Outdoor Set up:

- ☐ Clean-up campaigns to promote better waste management in schools and communities.
- ☐ Tree planting to encourage environmental conservation.
- ☐ Field visits to recycling plants or waste management facilities.
- ☐ Interviews with community leaders or environmental experts about resource management efforts.
- ☐ Awareness campaigns where students create posters or present skits on responsible resource use.

**Content Tips**

People can engage in actions that help in addressing Challenges of Rapid Population Growth. The activities could include;

- ☐ Improving Waste Management: Encouraging recycling, proper disposal of waste, and community clean-up initiatives.
- ☐ Efficient Resource Management: Promoting sustainable use of water, energy, and land to ensure availability for future generations.
- ☐ Raising Awareness: Educating communities about family planning, responsible resource use, and sustainable living practices.
- ☐ Policy Advocacy: Engaging in discussions about government policies and suggesting solutions to manage population growth effectively.
- ☐ Encourage learners to participate in the implementation of actions within their communities aimed at addressing challenges faced by rapid population growth.

Teaching and Learning Materials

- ☐ Posters and infographics on waste management and resource conservation.
- ☐ Videos/documentaries on sustainable development practices.
- ☐ Articles and reports on Zambia's environmental challenges and solutions.
- ☐ Recycling bins and materials for practical demonstrations.
- ☐ Interview guide for survey and interviews.

Expected Standard

- ☐ Challenges of rapid population growth addressed accordingly



Assessment

- ☐ Observe group discussions on proposed solutions to address the challenges of rapid population growth.
- ☐ Administer essays on the importance of sustainable practices in managing population growth.
- ☐ Assign community project where students apply waste management, conservation, or awareness strategies and present their outcomes.

2.1.4. SUB-TOPIC 4: MIGRATION

Introduction

Migration refers to the movement of people from one place to another, often in search of better living conditions, employment opportunities, or safety. In Zambia, migration occurs both within the country (internal migration) and across national borders (international migration). Internal migration includes rural-to-urban movement, seasonal migration for agricultural work, and relocation due to economic factors. Understanding migration helps in analyzing its causes, effects, and potential solutions to related challenges.

Specific Competence (s) - learners to:

Apply knowledge of migration in aspects of daily life



Activity 1

Discussing Concepts of Migration (migration, immigration, emigration) internal (rural-urban, seasonal and international migrations...)

Learning Environment

The learning environment for this activity would be indoor set up

Indoor Set up:

- ☐ Learners in groups discuss concepts on migration.
- ☐ Watching videos or presentations on different types of migration.
- ☐ Creating a concept map showing types of migration (internal vs. international).



Content Tips

- ☐ Migration: the movement of people from one place to another permanently or temporally.
- ☐ Immigration: when people enter a new country to live permanently.
- ☐ Emigration: when people leave their home country to settle elsewhere.
- ☐ Internal migration: movement of people within a country.
- ☐ Rural-Urban migration: people moving from villages to cities for better opportunities.
- ☐ Urban-Rural migration: people moving from urban areas to rural for various reasons.
- ☐ International migration: movement across national borders, often for employment or education.
- ☐ Circular migration: temporary movement of people between two or more locations often for work or family reasons.
- ☐ Seasonal migration: is the temporary movement of people from one region to another in response to changes in season, weather patterns or agricultural cycles.
- ☐ Voluntary migration: migration that occurs by choice often for economic or

educational reasons.

- ☐ Forced migration: migration that occur due to coercion, conflict or persecution.
- ☐ Asylum seeker: a person who seeks protection from another country due to persecution or conflicts.

Teaching and Learning Materials

- ☐ Maps of Zambia and the World to illustrate migration patterns.
- ☐ Videos/documentaries on migration.
- ☐ Case studies and reports on migration trends in Zambia.
- ☐ Charts and diagrams showing migration types.



Assessment

- ☐ Facilitate class discussions on migration concepts.
- ☐ Observe group presentations on different types of migration.



Activity 2

Analysing the Causes of Migration (employment, social services and recreational facilities, disputes, natural disasters, food insecurity...)

Learning Environment

The learning environment for this activity would be both indoor and outdoor set up

Indoor Set up:

- ☐ Learners in groups to discuss on causes of migration.
- ☐ Group work where students categorize different migration causes.
- ☐ Watching videos or documentaries on causes of migration.

Outdoor Set up:

- ☐ Conducting surveys within the community to identify reasons people migrate.
- ☐ Interviews with migrants about their reasons for migrating.
- ☐ Role-playing activities demonstrating different migration scenarios.



Content Tips

There are many causes of migration. The causes are divided into push and pull factors.

- ☐ Push factors are those things that force people to migrate while pull factors are those that attract people to settle in different areas.
- ☐ Employment Opportunities: People move to areas with better job prospects, such as cities or foreign countries.
- ☐ Access to Social Services: Migration occurs when people seek better healthcare, education, and housing.
- ☐ Recreational and Urban Facilities: Cities attract migrants with entertainment centers, modern infrastructure, and advanced services.
- ☐ Conflicts and Disputes: Wars, political instability, and ethnic conflicts force people to migrate for safety.
- ☐ Natural Disasters: Floods, droughts, and other disasters push people to safer areas.
- ☐ Food Insecurity: When agricultural production is low, people migrate to regions

with more stable food supplies.

Teaching and Learning Materials

- ☐ News articles and reports on migration causes.
- ☐ Documentaries and case studies on migration.
- ☐ Interview guides for conducting an investigation.
- ☐ Charts showing push and pull factors of migration.



Assessment

- ☐ Facilitate class discussions on observed migration causes.
- ☐ Administer essays on the causes of migration
- ☐ Assign research projects to analyse specific migration causes in Zambia



Activity 3

Examining the effects of Migration on the destination area and the place of origin (receiving destination: positive and negative; place of origin: positive and negative)

Learning Environment

The learning environment would involve both indoor and outdoor set up

Indoor Set up:

- ☐ Learners in groups discuss effects of migration for both origin and destination areas.
- ☐ Group work where learners categorize positive and negative effects.
- ☐ Watching videos or documentaries on migration impacts.
- ☐ Learners debate on whether migration is more beneficial or not.

Outdoor Set up:

- ☐ Interviews with local business owners about the impact of migration on the workforce.
- ☐ Field visits to urban centres experiencing high migration to observe housing and service demand.
- ☐ Role-playing exercises where students act as migrants and community members discussing effects of migration on destination and place of origin areas.



Content Tips

Migration has both positive and negative effects on both destination and place of origin.

- ☐ Positive effects on the Destination Area:
 - ✓ Cultural Diversity: New ideas, traditions, and skills enrich the community.
 - ✓ Labour Force Growth: Migrants contribute to economic activities and business expansion.
 - ✓ Increased Economic Activities: More workers and consumers stimulate business and trade.
- ☐ Negative effects on the Destination Area:
 - ✓ Strain on Public Resources: Increased demand for healthcare, education, and infrastructure.
 - ✓ Pressure on Housing: Migration can lead to overcrowding and high housing costs.

- ✓ Cultural and Social Tension: Conflicts may arise due to differences in culture and competition for jobs.
- ❑ Positive effects on the area of origin
 - ✓ Reduced Unemployment: Fewer people competing for limited jobs.
 - ✓ Relief from Population Pressure: Less demand on local resources.
- ❑ Negative effects on the area of origin
 - ✓ Brain Drain: Loss of skilled workers, reducing development potential.
 - ✓ Labour Shortages: Fewer workers in key sectors like agriculture and healthcare.
 - ✓ Economic Stagnation: Less workforce leads to slower economic growth.
 - ✓ Reduced tax revenue: fewer workers in local businesses like shops leads to low tax collection

Teaching and Learning Materials

- ❑ Maps to illustrate migration trends.
- ❑ News articles and reports on migration effects.
- ❑ Documentaries or videos on migration.
- ❑ Interview guides on conducting interview studies.
- ❑ Charts comparing positive and negative migration effects



Assessment

- ❑ Assign research projects on how migration has shaped specific communities.
- ❑ Coordinate group discussions on observed migration effects.
- ❑ Facilitate group presentations explaining how migration affects different regions.
- ❑ Assign essays analyzing migration's impact on Zambia.



Activity 4

Relating migration to various aspects of life. (*workforce, cultural and societal changes, environmental issues in the area of origin and receiving destination...*)

Learning Environment

The learning environment for this activity would be both indoor and outdoor set up

Indoor Set up:

- ❑ In groups learners to analyze migration's impact on work, culture and the environment.
- ❑ Watching videos or documentaries on migration and its effects.
- ❑ Debates on whether migration benefits the society or not.

Outdoor Set up:

- ❑ Interviews with professionals and business owners on how migration affects the workforce.
- ❑ Field visits to urban and rural areas to observe migration-related environmental changes.
- ❑ Studies through interviews within the community on the effects of migration on various aspects of life.
- ❑ Role-playing exercises where learners explore different views on migration.



Content Tips

Migration is related to various aspects of life in different ways. These include;

- ☐ Workforce
 - ✓ Migration helps fill labour gaps in industries, agriculture, and services.
 - ✓ Migrants may face job competition, leading to employment challenges.
- ☐ Cultural and Societal Changes
 - ✓ Migrants introduce new languages, traditions, and foods, enriching local culture.
 - ✓ Migration can lead to cultural tensions, erosion or integration difficulties.
- ☐ Environmental Issues
 - ✓ Urban migration contributes to deforestation, pollution, and waste accumulation.
 - ✓ Rural areas may experience land neglect or reduced agricultural activity.

Teaching and Learning Materials

- ☐ News articles and reports on migration's social and economic impact.
- ☐ Documentaries on migration and its environmental effects.
- ☐ Interview guides for study interviews.
- ☐ Charts comparing migration's effects on different aspects of life.

Expected standard

Knowledge of migration in aspects of daily life applied accordingly.



Assessment

- ☐ Ask learners to;
 - ✓ discuss the impacts of migration on various aspects of life
 - ✓ present on specific effects of migration.
 - ✓ compose essays on the influence of migration on Zambia's workforce, culture, or environment.
 - ✓ conduct research projects on how migration has changed specific communities.



Summary

Understanding of Population composition

- ☐ Zambia's population composition is characterized by a youthful age structure with almost a balanced sex ratio and majority of people living in rural areas.
- ☐ While the population is predominantly of African descent, there are also minority racial and ethnic groups, including people from Europe, Asia, and mixed heritage particularly in urban areas.
- ☐ These diverse groups contribute to the country's cultural and economic landscape. The composition presents both opportunities and challenges such as need for increased investment in Education, health care and employment to support the growing young population.
- ☐ Understanding these demographic patterns such as age, gender and ethnic diversity is essential for effective national planning and sustainable development in the country.

Analyse Population distribution

- ❑ Zambia's population is unevenly distributed across its ten (10) Provinces. It has sparsely, moderately and densely populated areas.
- ❑ The country's population distribution is influenced by various factors such as; climate, relief, soils, rainfall distribution, economy activities, Government policies, transport etc.
- ❑ Zambia has experienced significant population growth due to factors such as high birth rates, declining death rates, early marriages, and immigration.
- ❑ Understanding population distribution is crucial for effective planning, resource allocation and decision making in various sectors, such as healthcare, education and infrastructure development.

Challenges of Rapid population growth

- ❑ Zambia's rapid population growth increase has led to both challenges and opportunities.
- ❑ The country's rapid population growth has been caused by high birth rates, early marriage, declining death rates, better health care services, education etc
- ❑ Negative effects of rapid population growth include unemployment, overcrowding, increased crime rates, and pressure on social services like healthcare and education.
- ❑ However, population growth also brings benefits, such as a larger workforce, better utilization of natural resources, and improved security.
- ❑ Managing these challenges through effective policies, waste management, resource conservation, and public awareness is crucial for sustainable development.

Knowledge of Migration in aspects of daily life

- ❑ Knowledge of migration in aspects of daily life terms such as immigration, emigration, and migration are fundamental concepts in people's lives.
- ❑ By exploring these concepts, learners gain an understanding of the different types of movement people experience, both within Zambia and across borders.
- ❑ Migration is caused by push and pull factors influencing due to factors such as economic reasons, safety concerns, or the search for better living conditions.
- ❑ Migration have both positive and negative effects on the destination area and the place of origin are both such as cultural diversity, a growing labor force, and economic growth, while negative effects involve pressure on resources and housing. Migration on the place of origin can reduce unemployment but also lead to labor shortages and brain drain.
- ❑ Migration further affects the workforce, cultural and societal changes, and environmental conditions resulting in a diverse labor force, cultural exchange, and societal tensions, while also influencing environmental changes like urbanization and deforestation.

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SAMPLE LESSON PLAN**SCHOOL:** MAKUMBI SECONDARY SCHOOL**NAME OF TEACHER:** GEO**DATE:** 21/ 04/ 2025**CLASS:** Form 1 B**DURATION:** 80 Minutes**TIME:** 08:20-09:40**TOTAL NO. OF PUPILS:** 50**SUBJECT:** Geography**TOPIC:** 1.4. Natural Resources (*This must come from the syllabus with its serial number*)**SUB-TOPIC:** 1.4.1 Natural Resources; (*This must come from the syllabus with its serial number*)**GENERAL COMPETENCES:** Analytical thinking, Citizenship, Collaboration, and Environmental Sustainability.**SPECIFIC COMPETENCES:** 1.4.1.1 Use Renewable and non-Renewable resources responsibly. (*This must come from the syllabus with its serial number*)**ACTIVITY 1:** Identifying different Natural Resources**LESSON GOAL:** To enable learners, appreciate the importance of natural resources and develop the ability to use them responsibly and ensuring sustainable development.**RATIONALE:** This lesson is on **natural resources** and is particularly focused on their identification. Through group discussions and collaboration, learners will acquire a value of appreciating natural resources. This is Lesson 1 of 3 learning activities.**PRIOR KNOWLEDGE:** learners have knowledge of natural resources from the local environment.**REFERENCES:** - Mashekwa. J (2022). Book World Africa: Senior Secondary School Geography
Learner's Book Grade.12

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Geography

LEARNING ENVIRONMENT**Indoor Set up:** classroom (Teacher/learners)**Outdoor Set up:** local immediate environment.**Technological environment:** Laptops, tablets, smart phones**TEACHING AND LEARNING MATERIALS/RESOURCES**

- ☐ Note books
- ☐ Pictures and videos of natural resources
- ☐ Samples of some natural resources

EXPECTED STANDARD: Renewable and non-renewable resources used responsibly. (*This must come from the syllabus as it is*)**LESSON PROGRESSION:**

STAGES	TEACHER'S ROLE	LEARNERS' ROLE	ASSESSMENT CRITERIA
Introduction 5 Minutes	- Ask learners; 'Have you ever wondered where your desk, clothes and the food you eat (sandwich, nshima, bread) comes from? - Probe their' answers	Brainstorming on the sources of desks, clothes and food.	Sources of desks, clothes and food identified correctly.
Lesson Development 20 Minutes	- Divide learners into smaller groups and let them identify the natural resources they know. ✓ Display charts, pictures of natural resources ✓ Show videos of natural resources ✓ Consolidate learners' findings	- Learners in groups identify the natural resources they know. ✓ Present identified natural resources to the class	Natural resources identified correctly
25 Minutes	- Take learners outside to their local environment to; ✓ Identify the locally available natural resources	Identify the locally available natural resources	Locally available natural resources identified correctly.
10 Minutes	- Ask learners to classify the locally available natural resources into living and non-living. - Refine learners' findings	Classify the locally available natural resources into living and non-living.	Locally available natural resources classified into living and non-living appropriately.
Exercise	Teacher ask learners to explain the dangers of depleting the natural resources.	Learners to write an exercise explaining the dangers of depleting the natural resources.	Dangers of depleting natural resources explained clearly.
Homework	Task individual learners to prepare posters on how to avoid depletion of natural resources.	Individually, make and display posters on how to avoid depletion of natural resources around the school environment.	Posters made and displayed appropriately

Conclusion 10 Minutes	Recapitulate the lesson by consolidating the key points of the lesson.	Recapitulating the key points of the lesson to; a. Identify natural resources b. Classify natural resources	Key points of the lesson recapitulated accordingly.
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LESSON EVALUATION:

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